

Question 1 : Use Github Copilot to create a table SalesData whose columns are CustomerID, Name, Age, City, PurchaseAmount, PurchaseDate and then ask to insert 10,000 rows of random data.

Ans:-

```
SQLQuery3.sq...oy Das (76))*    SQLQuery3.sq...oy Das (53))*    SQLQuery2.s...oy Das (55)) ✕ ...  
1  create database SalesDB2026;  
2  go  
3  use SalesDB2026;  
4  
5  USE SalesDB2026;  
6  GO  
7  
8  CREATE TABLE SalesData  
9  (  
10 CustomerID INT IDENTITY(1,1) PRIMARY KEY,  
11 Name VARCHAR(50),  
12 Age INT,  
13 City VARCHAR(50),  
14 PurchaseAmount DECIMAL(10,2),  
15 PurchaseDate DATE  
16 );  
17 GO
```

```
SQLQuery3.sq...oy Das (76))*    SQLQuery3.s...y Das (53))* ✕  
1  USE SalesDB2026;  
2  GO  
3  SET NOCOUNT ON;  
4  
5  DECLARE @Counter INT = 1;  
6  
7  WHILE @Counter <= 10000  
8  BEGIN  
9      INSERT INTO SalesData  
10     (  
11         Name,  
12         Age,  
13         City,  
14         PurchaseAmount,  
15         PurchaseDate  
16     )  
17     VALUES  
18     (  
19         'Customer_' + CAST(@Counter AS VARCHAR(10)),  
20  
21         -- Random Age (18 to 60)  
22         FLOOR(RAND(CHECKSUM(NEWID())) * 43) + 18,  
23  
24         -- Random City  
25         CASE FLOOR(RAND(CHECKSUM(NEWID())) * 10)  
26             WHEN 0 THEN 'Kolkata'  
27             WHEN 1 THEN 'Delhi'  
28             WHEN 2 THEN 'Mumbai'  
29             WHEN 3 THEN 'Chennai'  
30             WHEN 4 THEN 'Bangalore'  
31             WHEN 5 THEN 'Hyderabad'  
32             WHEN 6 THEN 'Pune'  
33             WHEN 7 THEN 'Jaipur'  
34             WHEN 8 THEN 'Ahmedabad'  
35             ELSE 'Bhubaneswar'  
36         END,  
37  
38         -- Random Purchase Amount (100 to 5000)  
39         CAST((RAND(CHECKSUM(NEWID())) * 4900 + 100) AS DECIMAL(10,2)),  
40  
41         -- Random Purchase Date within the last 2 years  
42         DATEADD(  
43             DAY,  
44             -FLOOR(RAND(CHECKSUM(NEWID())) * 730),  
45             GETDATE()  
46         )  
47     );  
48  
49     SET @Counter = @Counter + 1;  
50  
51 END;  
52 GO  
  
60 %    No issues found  
  
Messages  
Commands completed successfully.  
Completion time: 2026-07-03T11:43:05.7652066+05:30
```

SQLQuery3.s...y Das (76))*	
1	SELECT COUNT(*) AS TotalRows
2	FROM SalesData;
100 % No issues found	
Results Messages	
TotalRows	
1	10000

Question 2 : Use Copilot to:

- Find total sales per city
- Top 5 cities by revenue

Ans:-

Find total sales per city

SQLQuery4.s...y Das (72))*	
1	-- Find Total Sales Per City
2	SELECT
3	City,
4	SUM(PurchaseAmount) AS TotalSales
5	FROM SalesData
6	GROUP BY City
7	ORDER BY TotalSales DESC;
100 % No issues found	
Results Messages	
City	TotalSales
1 Bhubaneswar	10077735.05
2 Kolkata	2632847.96
3 Delhi	2209102.77
4 Mumbai	2133834.81
5 Chennai	1780109.65
6 Bangalore	1703574.25
7 Hyderabad	1485421.85
8 Pune	1327465.30
9 Jaipur	1207565.57
10 Ahmedabad	1092984.20

Top 5 cities by revenue

SQLQuery5.s...y Das (91))* SQLQuery4.sq...oy Das (72))*

```
1  -- Top 5 Cities by Revenue
2  SELECT TOP 5
3      City,
4      SUM(PurchaseAmount) AS TotalRevenue
5  FROM SalesData
6  GROUP BY City
7  ORDER BY TotalRevenue DESC;
```

100 % No issues found

Results Messages

	City	TotalRevenue
1	Bhubaneswar	10077735.05
2	Kolkata	2632847.96
3	Delhi	2209102.77
4	Mumbai	2133834.81
5	Chennai	1780109.65

Question 3 : Ask Copilot:

- Find customers with purchases above average

Ans:-

SQLQuery1.sq...joy Das (82)) SQLQuery7.s...oy Das (80))

```
1  SELECT
2      CustomerID,
3      Name,
4      City,
5      PurchaseAmount
6  FROM SalesData
7  WHERE PurchaseAmount >
8      (
9          SELECT AVG(PurchaseAmount)
10         FROM SalesData
11      )
12  ORDER BY PurchaseAmount DESC;
```

100 % No issues found Ln: 12, Ch: 30 SPC CRLF UTF-8

Results Messages

	CustomerID	Name	City	PurchaseAmount
1	74	Customer_74	Bangalore	4999.62
2	9188	Customer_9188	Delhi	4999.33
3	8999	Customer_8999	Bhubaneswar	4999.04
4	2939	Customer_2939	Chennai	4998.30
5	8329	Customer_8329	Bhubaneswar	4997.48
6	6927	Customer_6927	Bangalore	4997.36
7	9333	Customer_9333	Bhubaneswar	4996.37
8	2423	Customer_2423	Hyderabad	4996.19
9	5153	Customer_5153	Mumbai	4995.32
10	2198	Customer_2198	Chennai	4995.02
11	9669	Customer_9669	Bhubaneswar	4994.14
12	2984	Customer_2984	Hyderabad	4994.12
13	3918	Customer_3918	Bhubaneswar	4993.60
14	5033	Customer_5033	Bhubaneswar	4993.17
15	2965	Customer_2965	Chennai	4993.05
16	8055	Customer_8055	Bhubaneswar	4992.96
17	2402	Customer_2402	Kolkata	4992.78

Query executed successfully. DESKTOP-ERBRDA2\SQLEXPRESS ... DESKTOP-ERBRDA2\Sanjoy... SalesDB2026 00:00:00 Row: 1, Col: 1 5,019 rows

GitHub Copilot Chat

Create a table named SalesData with col...

```
SELECT
    LEFT(CONVERT(varchar(36),
        (ABS(CHECKSUM(NEWID())) %
        (SELECT CityName FROM Citi
        CAST((ABS(CHECKSUM(NEWIDC
        DATEADD(day, - (ABS(CHECKS
        FROM Tally;

    COMMIT TRAN;
END TRY
BEGIN CATCH
    IF XACT_STATE() <> 0 ROLLBACK
    THROW;
END CATCH;
GO
```

Confirm execution and specify the target connection/ editor (or database name) before I run the insert.

Question 2 : Use Copilot to:

- Find total sales per city
- Top 5 cities by revenue

1 reference

SQLQuery5.sql F:\PW Assignment- 10 G...

Active Document

Agent GPT-5 mini

Dataset : Dummy dataset

Question 4 : Identify and handle missing values in the dataset using Copilot.

Identify Missing Values

1. Open Excel_dataset.csv in Excel.
2. Select the entire dataset (Ctrl + A).
3. Go to Home → Find & Select → Go To Special → Blanks.
4. Excel will highlight all blank cells.

The screenshot shows the Microsoft Excel interface with a dataset named 'Excel_dataset'. The dataset is displayed in a table with the following columns: Customer_Name, Region, Product, Sales, Quantity, and Order_Date. The 'Sales' column contains values like 'NA' and 'two', and the 'Quantity' column contains values like 'two'. The 'Region' and 'Product' columns also contain 'two' values. The 'Order_Date' column contains dates. The 'Sales' column has missing values (NA) in rows 6, 13, 17, 20, 22, 23, 24, 25, 26, 27, and 28. The 'Quantity' column has missing values (two) in rows 4, 5, 10, 11, 12, 14, 15, 16, 18, 19, 21, 22, 23, 24, 25, 26, 27, and 28. The 'Region' column has missing values (two) in rows 4, 5, 10, 11, 12, 14, 15, 16, 18, 19, 21, 22, 23, 24, 25, 26, 27, and 28. The 'Product' column has missing values (two) in rows 4, 5, 10, 11, 12, 14, 15, 16, 18, 19, 21, 22, 23, 24, 25, 26, 27, and 28.

Handle Missing Values

Option 1: Replace Text Blanks with "Unknown"

For columns like Customer_Name, Region, and Product:

1. Select the column.
2. Press Ctrl + H.
3. In Find what, leave it blank.
4. In Replace with, type:

Unknown

5. Click Replace All.

Option 2: Replace Numeric Blanks with Average

For Sales

1. Calculate the average:

=AVERAGE(D:D)

2. Copy the average value.
3. Select blank cells in the Sales column and paste the value.

For Quantity

=AVERAGE(E:E)

Replace blank cells with this average.

Option 3: Remove Rows with Missing Values

1. Select the table.
2. Go to Data → Filter.
3. Filter (Blanks).
4. Select those rows.
5. Right-click → Delete Row.

Formula Bar: =IF(COUNTIFS(\$A:\$A,A2,\$B:\$B,B2,\$C:\$C,C2,\$D:\$D,D2,\$E:\$E,E2,\$F:\$F,F2)>1,"Duplicate","Unique")

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	Customer_Name	Region	Product	Sales	Quantity	Order_Date												
2	Karan		Monitor		4	08-06-2023	Unique											
3	Riya	South	Monitor		2	18-03-2023												
4	Amit	West		300	two	01-07-2023												
5	Sneha		Tablet	300	4	01-04-2023												
6	Sneha	West	Phone	NA	4	15-05-2023												
7	Sneha	South	Laptop	300		01-05-2023												
8	Rahul	South		400	4	11-05-2023												
9	Sneha	South	Phone	400		04-06-2023												
10	Riya	North	Monitor	400	1	27-08-2023												
11	Priya	West	Laptop	400	4	09-10-2023												
12	Karan	North	Monitor	100	two	14-01-2023												
13			Tablet	100	4	08-08-2023												
14	Amit	South	Phone	NA	4	04-03-2023												
15	Amit		Laptop	100		27-02-2023												
16	Karan		Laptop	200	4	21-10-2023												
17	Priya	North	Monitor	100	3	18-04-2023												
18	Rahul	West		NA	1	22-11-2023												
19		East	Phone	200	2	02-11-2023												
20		South	Laptop		two	26-01-2023												
21	Amit		Laptop	100	2	26-09-2023												
22	Rahul		Tablet	300		22-02-2023												
23	Riya		Phone	NA	two	23-11-2023												
24	Rahul	North	Phone	400	1	11-09-2023												
25	Riya	North	Laptop	200	1	15-03-2023												
26	Amit	East	Phone	NA	2	14-03-2023												
27	Rahul		Phone			25-05-2023												
28	Karan	North	Laptop	100	3	18-12-2023												

Excel_dataset

Question 5 : Task –

- Identify duplicate rows
- Remove duplicates using Copilot suggestion

Ans:-

Method 1: Identify Duplicate Rows in Excel

Using Conditional Formatting

1. Select the entire dataset.
2. Go to Home → Conditional Formatting → Highlight Cells Rules → Duplicate Values.
3. Choose a color and click OK.

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08-06-2023

	A	B	C	D	E	F	G
1	Customer_Name	Region	Product	Sales	Quantity	Order_Date	
2	Karan		Monitor		4	08-06-2023	
3	Riya	South	Monitor		2	18-03-2023	
4	Amit	West		300	two	01-07-2023	
5	Sneha		Tablet	300	4	01-04-2023	
6	Sneha	West	Phone	NA	4	15-05-2023	
7	Sneha	South	Laptop	300		01-05-2023	
8	Rahul	South		400	4	11-05-2023	
9	Sneha	South	Phone	400		04-06-2023	
10	Riya	North	Monitor	400	1	27-08-2023	
11	Priya	West	Laptop	400	4	09-10-2023	
12	Karan	North	Monitor	100	two	14-01-2023	
13			Tablet	100	4	08-08-2023	
14	Amit	South	Phone	NA	4	04-03-2023	
15	Amit		Laptop	100		27-02-2023	
16	Karan		Laptop	200	4	21-10-2023	
17	Priya	North	Monitor	100	3	18-04-2023	
18	Rahul	West		NA	1	22-11-2023	
19		East	Phone	200	2	02-11-2023	
20		South	Laptop		two	26-01-2023	
21	Amit		Laptop	100	2	26-09-2023	
22	Rahul		Tablet	300		22-02-2023	
23	Riya		Phone	NA	two	23-11-2023	
24	Rahul	North	Phone	400	1	11-09-2023	
25	Riya	North	Laptop	200	1	15-03-2023	
26	Amit	East	Phone	NA	2	14-03-2023	
27	Rahul		Phone			25-05-2023	
28	Karan	North	Laptop	100	3	18-12-2023	

Method 2: Using a Formula (to identify duplicate rows)

Suppose your data is in columns A:F and starts from row 2.

In a new column (G2), enter:

=COUNTIFS(\$A:\$A,A2,\$B:\$B,B2,\$C:\$C,C2,\$D:\$D,D2,\$E:\$E,E2,\$F:\$F,F2)>1

Press Enter and drag the formula down.

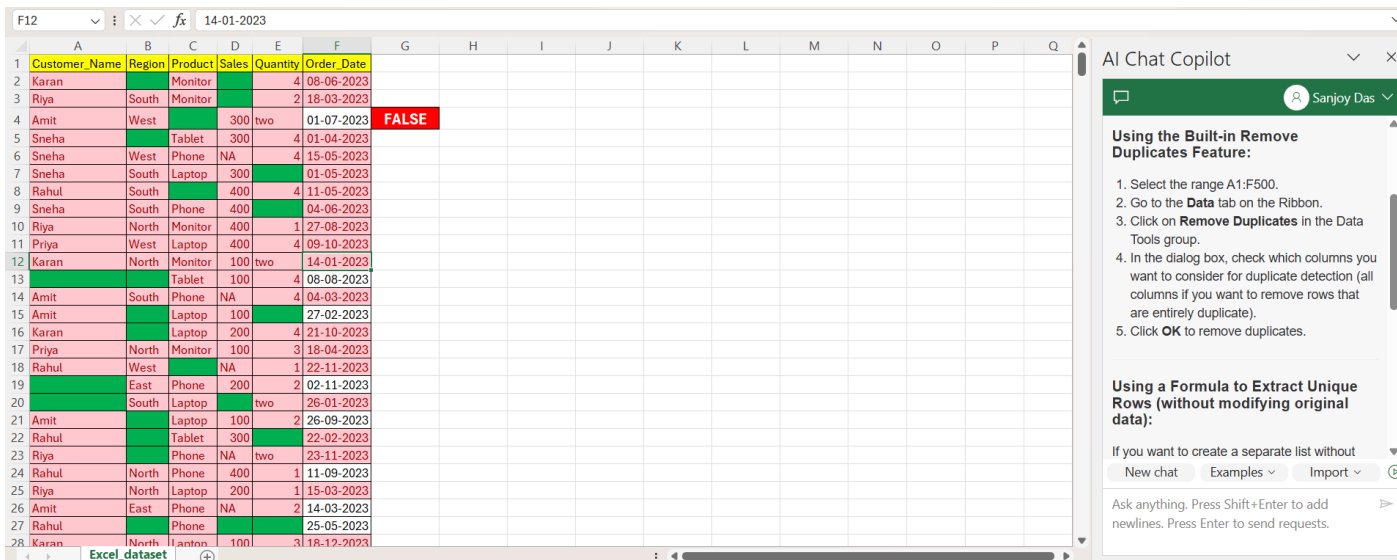
- **TRUE** = Duplicate row
- **FALSE** = Unique row

G4				fx		=COUNTIFS(\$A:\$A,A2,\$B:\$B,B2,\$C:\$C,C2,\$D:\$D,D2,\$E:\$E,E2,\$F:\$F,F2)>1				
	A	B	C	D	E	F	G	H	I	J
1	Customer_Name	Region	Product	Sales	Quantity	Order_Date				
2	Karan		Monitor		4	08-06-2023				
3	Riya	South	Monitor		2	18-03-2023				
4	Amit	West		300	two	01-07-2023	FALSE			
5	Sneha		Tablet	300	4	01-04-2023				
6	Sneha	West	Phone	NA	4	15-05-2023				
7	Sneha	South	Laptop	300		01-05-2023				
8	Rahul	South		400	4	11-05-2023				
9	Sneha	South	Phone	400		04-06-2023				
10	Riya	North	Monitor	400	1	27-08-2023				
11	Priya	West	Laptop	400	4	09-10-2023				
12	Karan	North	Monitor	100	two	14-01-2023				
13			Tablet	100	4	08-08-2023				
14	Amit	South	Phone	NA	4	04-03-2023				
15	Amit		Laptop	100		27-02-2023				
16	Karan		Laptop	200	4	21-10-2023				
17	Priya	North	Monitor	100	3	18-04-2023				
18	Rahul	West		NA	1	22-11-2023				
19		East	Phone	200	2	02-11-2023				
20		South	Laptop		two	26-01-2023				
21	Amit		Laptop	100	2	26-09-2023				
22	Rahul		Tablet	300		22-02-2023				
23	Riya		Phone	NA	two	23-11-2023				
24	Rahul	North	Phone	400	1	11-09-2023				
25	Riya	North	Laptop	200	1	15-03-2023				
26	Amit	East	Phone	NA	2	14-03-2023				
27	Rahul		Phone			25-05-2023				

Remove duplicates using Copilot suggestion

Method 1: Remove Duplicate Rows

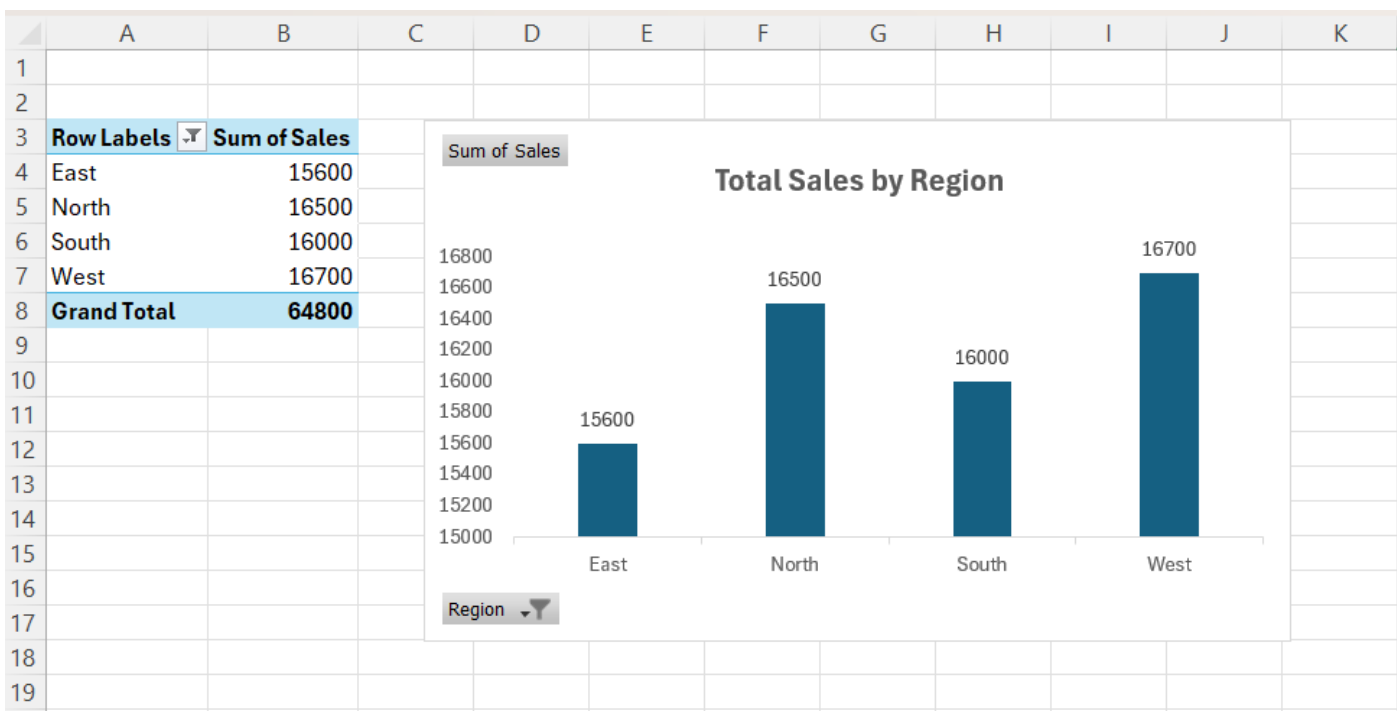
1. Select the entire dataset.
2. Go to the Data tab.
3. Click Remove Duplicates.
4. In the dialog box:
 - Check My data has headers.
 - Select all columns.
5. Click **OK**.



Question 6 : Generate:

- Column chart OR bar chart and Interpret which region has highest sales

Ans:-



Interpret the Chart

- West has the highest sales, with a total of 16,700.
- North is the second-highest performing region, with 16,500 in sales.
- South generated 16,000 in sales.
- East has the lowest sales, with 15,600.

Question 7 : Create a pivot table showing total sales by region and product.

Ans:-

Step 1: Select the Dataset

1. Open your Excel file.
2. Click on any cell inside the dataset.
3. Press Ctrl + A to select the entire table.

Step 2: Insert a Pivot Table

1. Go to the Insert tab.
2. Click PivotTable.
3. In the dialog box:
 - o Verify that the correct data range is selected.
 - o Choose New Worksheet.
4. Click OK.

Step 3: Arrange the Pivot Table Fields

In the PivotTable Fields pane:

1. Drag Region to the Rows area.
2. Drag Product to the Columns area.
3. Drag Sales to the Values area.

Excel will automatically display Sum of Sales.

Step 4: Format the Sales Values (Optional)

1. Right-click any value in the PivotTable.
2. Select Value Field Settings.
3. Click Number Format.
4. Choose Currency or Number and click OK.

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	A	B	C	D	E	F	G
1							
2							
3	Sum of Sales	Column Labels					
4	Row Labels	Laptop	Monitor	Phone	Tablet	(blank)	Grand Total
5	East	₹ 1,900.00	₹ 4,900.00	₹ 3,100.00	₹ 2,700.00	₹ 3,000.00	₹ 15,600.00
6	North	₹ 2,400.00	₹ 4,600.00	₹ 3,400.00	₹ 2,900.00	₹ 3,200.00	₹ 16,500.00
7	South	₹ 4,100.00	₹ 2,900.00	₹ 2,900.00	₹ 3,500.00	₹ 2,600.00	₹ 16,000.00
8	West	₹ 4,300.00	₹ 3,600.00	₹ 5,600.00	₹ 2,200.00	₹ 1,000.00	₹ 16,700.00
9	(blank)	₹ 7,300.00	₹ 6,800.00	₹ 5,600.00	₹ 6,500.00	₹ 6,700.00	₹ 32,900.00
10	Grand Total	₹ 20,000.00	₹ 22,800.00	₹ 20,600.00	₹ 17,800.00	₹ 16,500.00	₹ 97,700.00
11							

Insight:

- Monitor is the best-selling product with total sales of ₹22,800.
- Tablet has the lowest sales among the named products (₹17,800).
- Missing product values contribute ₹16,500 in sales.

Data Quality Insights

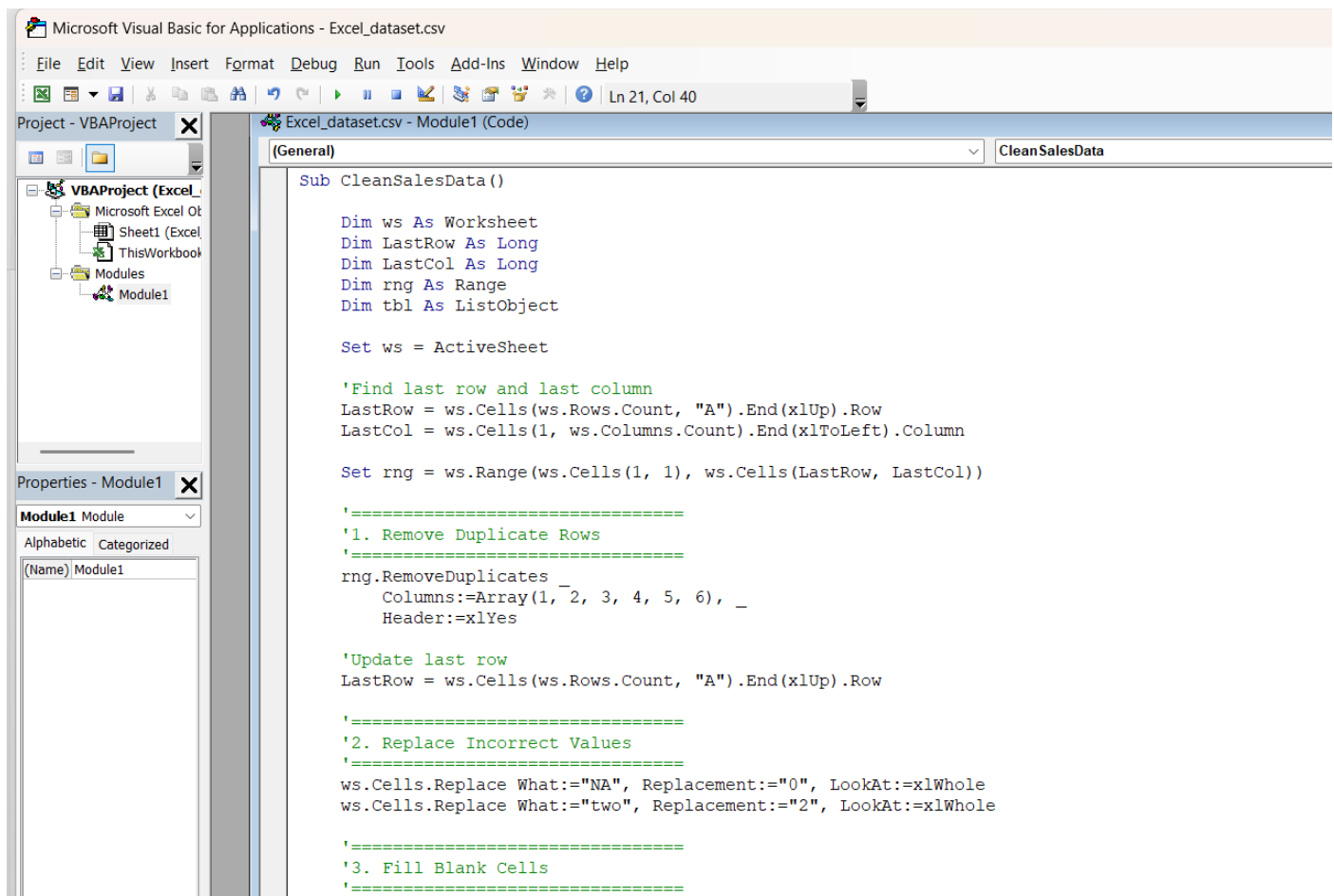
- 51 duplicate rows were found in the dataset.
- Several columns contain missing values.
- Text values such as "NA" and "two" were present in numeric columns and needed to be cleaned before analysis.

Business Insights

1. West region generates the highest sales among the identified regions.
2. Monitor is the top-selling product and contributes the highest revenue.
3. Data quality issues (missing values and duplicates) can affect business decisions and should be addressed before further analysis.
4. Sales are relatively balanced across the four regions, with only a small difference between the highest and lowest performing regions.

Question 10 : Generate VBA code using Copilot for cleaning data (removing duplicates, filling null values, format table).

Ans:-



```

Microsoft Visual Basic for Applications - Excel_dataset.csv
File Edit View Insert Format Debug Run Tools Add-Ins Window Help
Ln 21, Col 40

Project - VBAProject
  VBAProject (Excel_dataset.csv)
    Microsoft Excel Worksheet
    Sheet1 (Excel_dataset.csv)
    ThisWorkbook
    Modules
      Module1

Properties - Module1
Module1 Module
  (Name) Module1

Excel_dataset.csv - Module1 (Code)
(General) CleanSalesData

Sub CleanSalesData()

    Dim ws As Worksheet
    Dim LastRow As Long
    Dim LastCol As Long
    Dim rng As Range
    Dim tbl As ListObject

    Set ws = ActiveSheet

    'Find last row and last column
    LastRow = ws.Cells(ws.Rows.Count, "A").End(xlUp).Row
    LastCol = ws.Cells(1, ws.Columns.Count).End(xlToLeft).Column

    Set rng = ws.Range(ws.Cells(1, 1), ws.Cells(LastRow, LastCol))

    '=====
    '1. Remove Duplicate Rows
    '=====
    rng.RemoveDuplicates _
        Columns:=Array(1, 2, 3, 4, 5, 6), _
        Header:=xlYes

    'Update last row
    LastRow = ws.Cells(ws.Rows.Count, "A").End(xlUp).Row

    '=====
    '2. Replace Incorrect Values
    '=====
    ws.Cells.Replace What:="NA", Replacement:="0", LookAt:=xlWhole
    ws.Cells.Replace What:="two", Replacement:="2", LookAt:=xlWhole

    '=====
    '3. Fill Blank Cells
    '=====
  
```

